

Building a conceptual 3D geological model of the Wollaston fold-thrust belt, northern Saskatchewan: Implications for pegmatite-related rare earth elements (REE) targeting

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Introduction

Rare Earth Elements (REE) are essential for renewable technologies [1], but pegmatite-hosted REE systems remain difficult to explore because they are commonly small, structurally controlled [2], and poorly constrained in 3D [3, 4]. This study forms part of the broader “Unlocking Canada’s Rare Earth Element (REE) Potential” initiative and represents one component of a PhD project focused on integrated geological–geophysical modelling of the Wollaston Domain. This domain is a prospective Precambrian terrane where regional lithotectonic architecture, structural corridors, and known REE occurrences provide a basis for improving regional-scale exploration models [5].

Objective

This study builds a regional-scale conceptual 3D geological model of the Wollaston Domain by integrating geological, geophysical, structural, and geochemical datasets to assess existing interpretations, identify knowledge gaps, and refine preliminary criteria for REE pegmatite

Research Hypotheses

H1: REE-bearing pegmatites may reflect the interaction between fertile crustal source regions and inherited structural anisotropy.

H2: NE-trending D1/D2 ductile structures may have acted as melt–fluid pathways, whereas NW-trending brittle faults (possibly late D3) may have influenced localization or preservation.

H3: Integrating structural patterns with radiometric responses may help identify higher-priority targets for REE pegmatite exploration.

Geological Setting

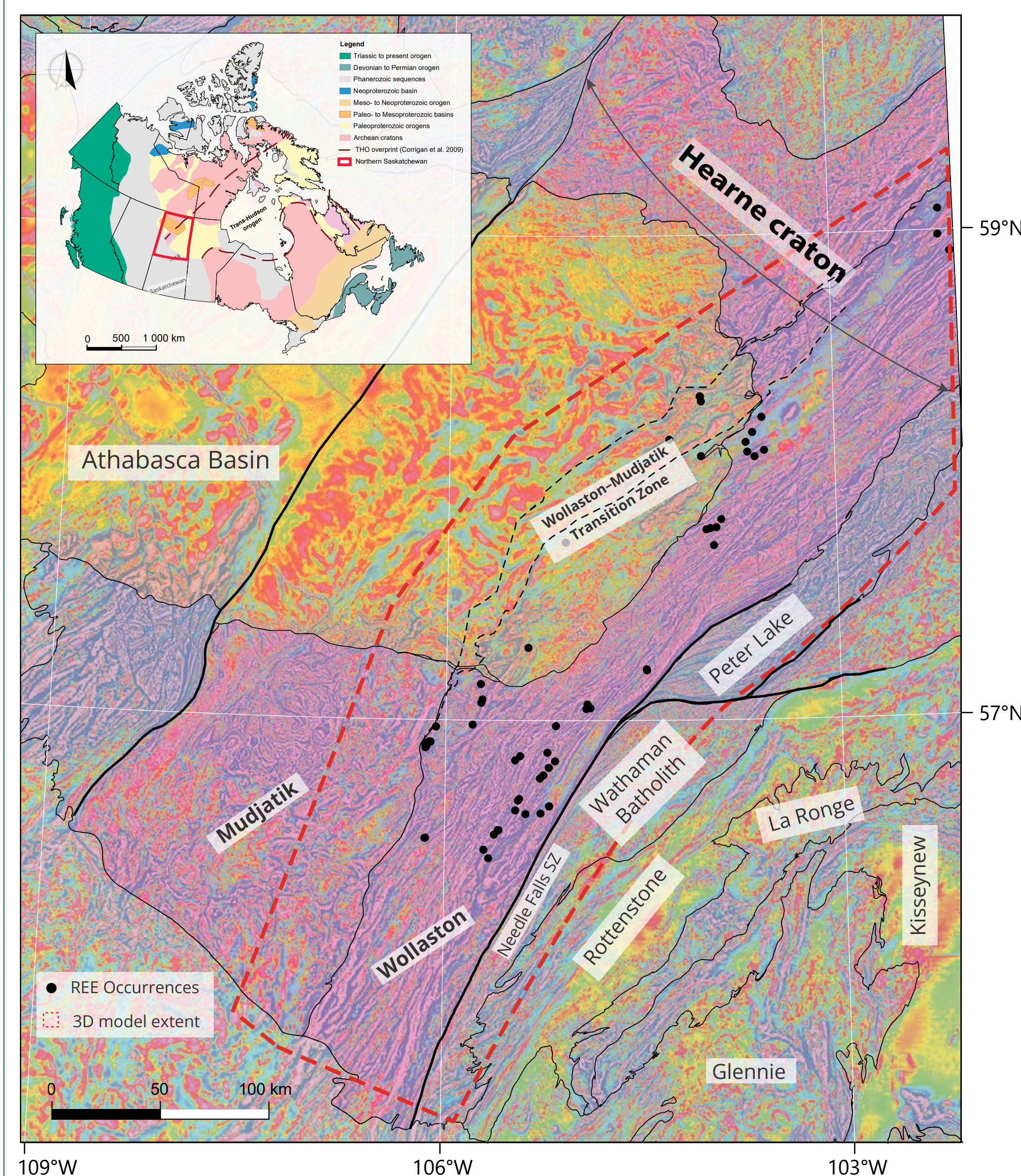


Figure 1. Location and geological–geophysical setting of the Wollaston Domain. The inset shows the study area within Canada and the Canadian Precambrian Shield. The main map shows lithotectonic domains of northern Saskatchewan [14] overlain on the first vertical derivative of total magnetic intensity (TMI)[15], with REE occurrences and the approximate 3D model extent [14].

- The Wollaston fold–thrust belt is a NE-trending structural corridor within the Hearne craton, developed during Paleoproterozoic Trans-Hudson orogenesis [6–10].
- The belt records crustal shortening, high-grade metamorphism, partial melting, and structural re-activation [6, 9, 10].
- Archean basement and Paleoproterozoic Wollaston Group rocks are interfolded within crustal-scale thrust sheets [6–10].
- The WMTZ separates basement-cored domes of the Mudjatik Domain from the linear fold–thrust architecture of the Wollaston Domain [6, 12].
- Graphitic and highly deformed rocks in the WMTZ may have acted as conductive fluid/melt pathways relevant to U and REE-bearing pegmatite systems [12, 13].

Data & Integration workflow

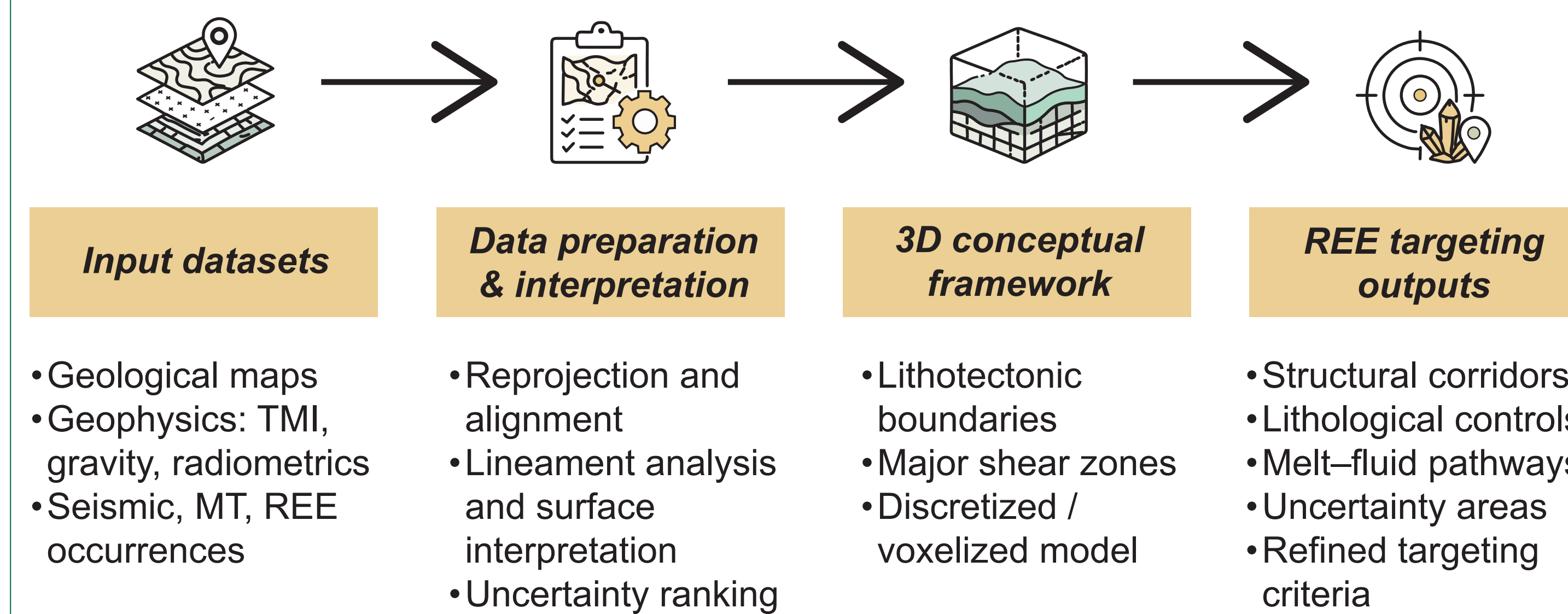


Figure 2. Integrated workflow used to build the 3D conceptual lithotectonic model. Surface geological and geophysical datasets are combined with subsurface seismic and MT constraints to evaluate structural architecture, uncertainty, and criteria relevant to REE pegmatite targeting.

3D Conceptual Model of the Wollaston Domain

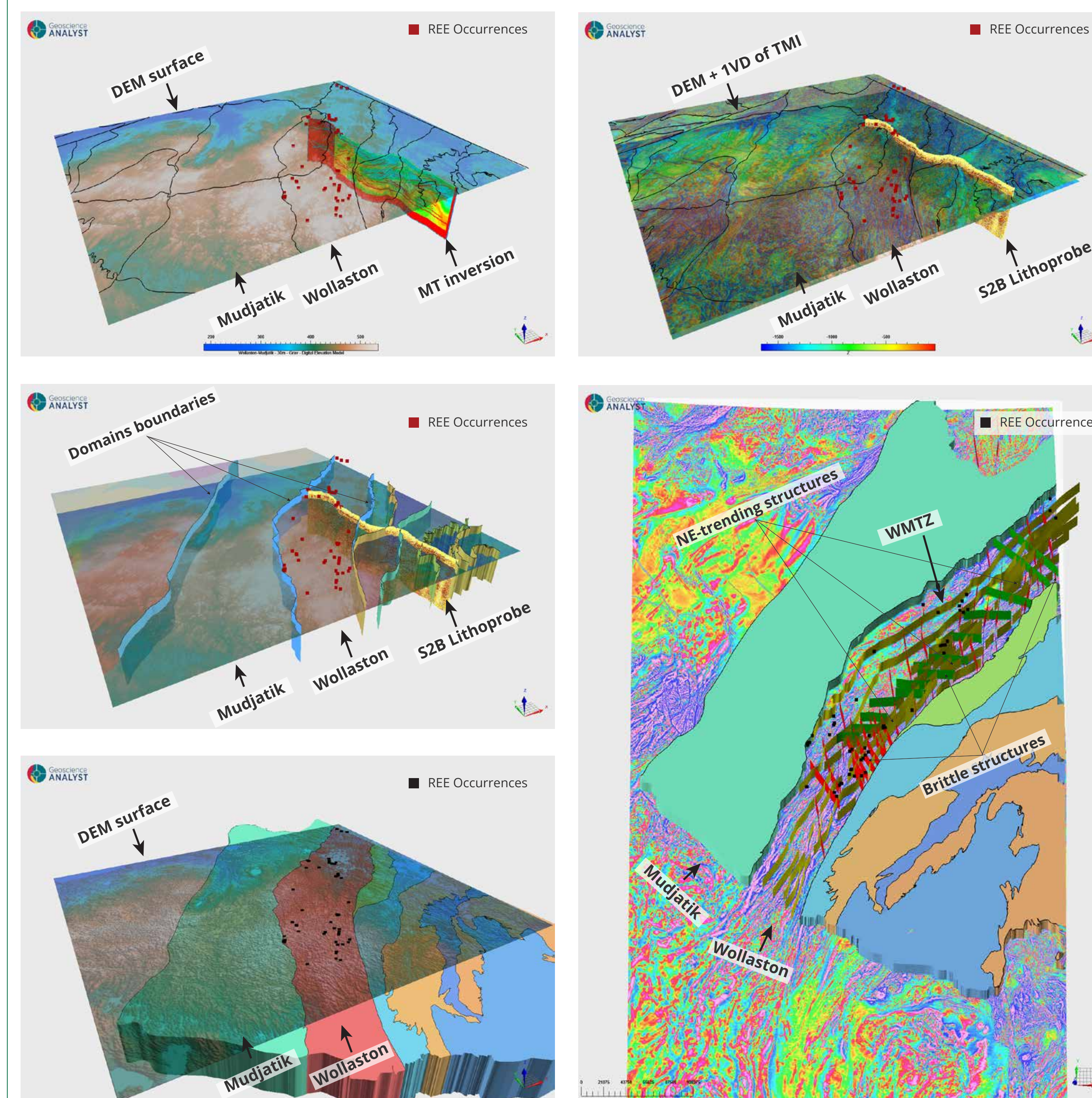


Figure 3. Preliminary 3D conceptual model inputs and derived structural framework. A) topography/DEM surface [16] with MT inversion interpretation [17]. B) Topography draped with the first vertical derivative of total magnetic intensity (TMI) and the Lithoprobe THOT S2B seismic transect [18], along which the MT profile was also acquired. C) Interpreted lithotectonic domain-boundary surfaces constrained by topography and the S2B transect. D) Preliminary lithotectonic block model of the Wollaston Domain. E) interpreted structural model, with lineament-derived faults overlain on the first vertical derivative of TMI.

Key Findings

- Surface geology, geophysics, and subsurface constraints were integrated into a preliminary regional lithotectonic framework.
- REE occurrences are spatially associated with NE-trending structural corridors and interpreted shear-zone surfaces.
- NW-trending brittle structures may record late reactivation relevant to pegmatite localization or preservation.
- eTh radiometric patterns help evaluate preliminary links between REE occurrences, structures, and lithological domains.
- The model highlights uncertainty areas and refines preliminary criteria for REE pegmatite targeting.

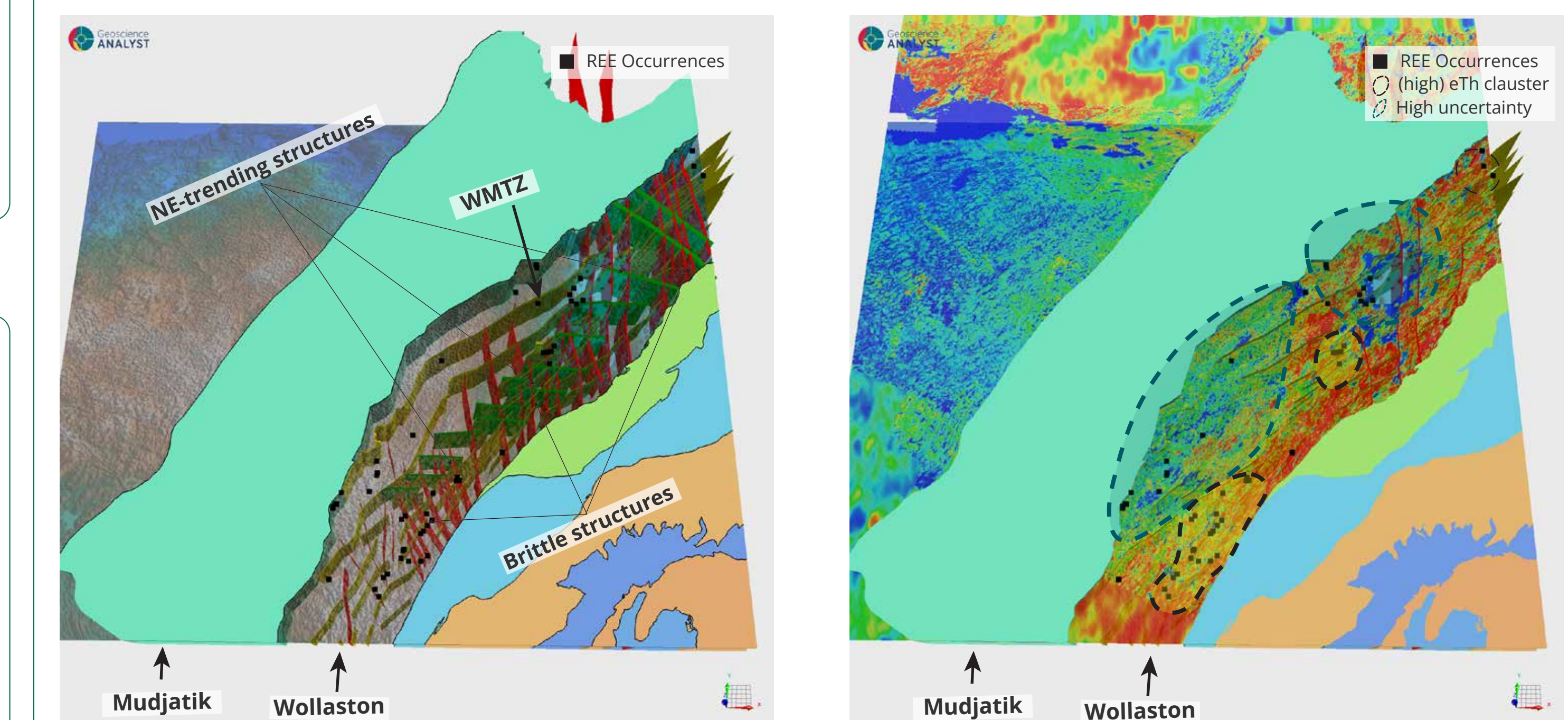


Figure 4. Preliminary spatial comparison between REE pegmatite occurrences, interpreted structures, and geophysical responses in the Wollaston Domain. (A) REE occurrences plotted with lineament-derived faults and shear-zone surfaces interpreted from the first vertical derivative of total magnetic intensity (TMI), highlighting the spatial relationship between pegmatite occurrences and NE–NW structural corridors. (B) Same structural framework overlain on equivalent thorium (eTh) radiometric data [15], showing the relationship between REE occurrences, interpreted structures, and radiometric response.

Future Work

- Refine the conceptual 3D lithotectonic model and uncertainty framework.
- Reinterpret Lithoprobe THOT Line S2B to better constrain subsurface geometry.
- Use petrophysical data for forward magnetic, gravity, and MT modelling.
- Perform geologically constrained inversions to iteratively improve the model.
- Evaluate refined criteria for preliminary REE pegmatite targeting.

Acknowledgements

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